

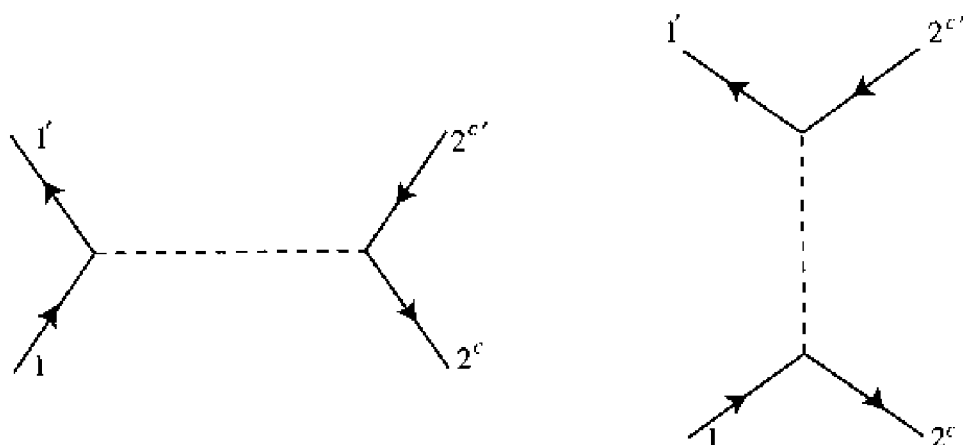


Figure 6.5. The connected second-order diagrams for fermion–antifermion scattering in a theory with interaction (6.1.18). Here straight lines represent fermions or antifermions, depending on the arrow direction; dotted lines are neutral bosons. There is again a minus sign difference in the contributions of these two diagrams, arising from an extra interchange of fermion operators in the pairings represented by the second diagram.

general Feynman diagrams form either chains of lines that pass through the diagram with arbitrary numbers of interactions with the boson fields, as in Figure 6.6, or else fermionic *loops*, like that shown in Figure 6.7. Consider the effect of adding a fermionic loop with M corners to the Feynman diagram for any process. This corresponds to the pairing of fermionic operators

$$[\psi(x_1)\bar{\psi}(x_2)] [\psi(x_2)\bar{\psi}(x_3)] \cdots [\psi(x_M)\bar{\psi}(x_1)] . \quad (6.1.25)$$

On the other hand, these operators appear in Eq. (6.1.1) in the order

$$\bar{\psi}(x_1)\psi(x_1)\bar{\psi}(x_2)\psi(x_2) \cdots \bar{\psi}(x_M)\psi(x_M) . \quad (6.1.26)$$

To go from (6.1.26) to (6.1.25) requires an odd permutation of fermionic operators (move $\bar{\psi}(x_1)$ to the right past $2M - 1$ operators) so the contribution of each such fermionic loop is accompanied with a minus sign.

These rules yield the full S -matrix, including contributions from processes in which various clusters of particles interact in widely separated regions of spacetime. As discussed in Chapter 4, to calculate the part of the S -matrix that excludes such contributions, we should include only *connected* Feynman diagrams. In particular, this excludes lines passing clean through the diagram without interacting, which would yield the factors (6.1.13).

To make the Feynman rules perfectly clear, we will calculate the low-order contributions to the S -matrix for particle scattering in two different theories.